CHAPTER

1

REAL NUMBER SYSTEM

1.1 Algebraic and Order axioms

The real number system consists of the real numbers, together with the two operations, addition (denoted by $+$) and multiplication (denoted by $\times$) and the less than relation (denoted by $<$). One also singles out two particular real numbers, zero or 0 and one or 1. If a and b are real numbers, then so are $a + b$ and $a \times b$. We say that the real numbers are closed under addition and multiplication. We usually write

$$ab \text{ for } a \times b.$$

For any two real numbers a and b, the statement $a < b$ is either true or false. We will soon see that one can define subtraction and division in terms of $+$ and $\times$; and $\leq, >$, etc. can be defined from $<$. There are three categories of properties of the real number system: the algebraic properties, the order properties and the completeness property. We will discuss the completeness property in a later section of this chapter. Here we begin with certain basic algebraic and order properties, usually called the algebraic and order axioms, from which we can prove all the other algebraic and order properties of the real numbers. For all real numbers a, b and c :

1. $(a+b)+c = a+(b+c)$ (associative axiom for addition)

2. $a+0 = 0+a = a$ (additive identity axiom)

3. there is a real number, denoted $-a$, such that $a+(-a) = (-a)+a = 0$ (additive inverse axiom)

4. $a+b = b+a$ (commutative axiom for addition)

5. $(a \times b) \times c = a \times (b \times c)$ (associative axiom for multiplication)

6. $a \times 1 = 1 \times a = a$, moreover $0 \neq 1$ (multiplicative identity axiom)

7. if $a \neq 0$ then there is a real number, denoted a^{-1}, such that $a \times a^{-1} = a^{-1} \times a = 1$ (multiplicative inverse axiom)

8. $a \times (b+c) = a \times b + a \times c$ (distributive axiom)

9. $a \times b = b \times a$ (commutative axiom for multiplication)

10. exactly one of the following holds: $a < b$, $a = b$ or $b < a$ (trichotomy axiom)

11. if $a < b$ and $b < c$, then $a < c$ (transitivity axiom)

12. if $a < b$ then $a + c < b + c$ (addition and order axiom)

13. if $a < b$ and $0 < c$, then $a \times c < b \times c$ (multiplication and order axiom)

1.1.1 Remarks

- For equality, denoted by the symbol "=", we mean "the same thing as", or equivalently, "is the same real number as". We take "=" to be a logical notion and do not write axioms for it. [2] Instead, we use any property of "=" which follow from its logical meaning. For example $a = a$; if $a = b$ then $b = a$; if $a = b$ and $b = c$ then $a = c$; if $a = b$ and something is true of a then it's also true of b (since a and b denote the same real number!).

- When we write $a \neq b$, we just mean that a is different real number as b.

- The assertion $0 \neq 1$ in axiom 6 may seem silly. But it doesn't follow from the other axioms, since all the other axioms hold for the set containing just the number 0 .

- Some of the axioms are redundant. For example, from axiom 4 and the property $a+0=a$ it follows that $0+a=a$. Similar comments apply to axiom 3 ; and because of axiom 6 to axioms 8 and 9.

1.1.2 Algebraic Structures

- Axioms $1,2,3,4$ allow $\mathbb{R}$ the structure of an additive Abelian group.
- Axioms $5,6,7$ allow $\mathbb{R}$ the structure of a multiplicative group.
- Axioms $1,...,8$ allow us to justify that $\mathbb{R}$ is a field.
- Commutativity of the multiplication operation $\times$ makes $\mathbb{R}$ a commutative field.

From axiom 8, we can write

$$a\left(b+(-b)\right)=0=ab+a\left(-b\right),$$

which means

$$-\left(ab\right)=a\left(-b\right)$$

same as above we obtain

$$-\left(ab\right)=\left(-a\right)b.$$

1.1.3 Algebraic consequences

Certain not so obvious "rules", such as "the product of minus times minus is plus" and the rule for adding two fractions, follow from the axioms. If we want the properties given by axioms 1-9 to be true for the real numbers (and we do), then there is no choice other than to have $(-a)(-b)=ab$ and $(a/c)+(b/d)=(ad+bc)/cd$ (see the following theorem). We won't emphasise the idea of making deductions from the axioms. Nonetheless, you should have some appreciation of the ideas involved, and thus you should work through a couple of proofs.

Theorem 1.1. *If a,b,c,d are real numbers and $c\neq 0,d\neq 0$ then*

1. *$ac=bc$ implies $a=b$.*

2. $a0 = 0$

3. $-(-a) = a$

4. $(c^{-1})^{-1} = c$

5. $(-1)a = -a$

6. $a(-b) = -(ab) = (-a)b$

7. $(-a) + (-b) = -(a + b)$

8. $(-a)(-b) = ab$

9. $(a/c)(b/d) = (ab)/(cd)$

10. $(a/c) + (b/d) = (ad + bc)/cd$

1.2 $\mathbb{R}$ is a totally ordered field

Order consequences

All the standard properties of inequalities for the real numbers follow from axioms 1-13.
More definitions:
One defines " $>$ ", " $\leq$ ” and " $\geq$ ” in terms of $<$ as

$$a > b \text{ if } b < a,$$
$$a \leq b \text{ if } (a < b \text{ or } a = b),$$
$$a \geq b \text{ if } (a > b \text{ or } a = b).$$

(Note that the statement $1 \leq 2$, although it isn't one we are likely to make, is indeed true. Why?)
We define $\sqrt{b}$, for $b \geq 0$, to be the number $c \geq 0$ such that $c^2 = b$. Similarly, if n is a natural number, then $\sqrt[n]{b}$ is the number $c \geq 0$ such that $c^n = b$. To prove such a number c always exists requires the "completeness axiom" (see later). To prove the uniqueness of such a number requires the "order axioms". If $0 < a$, we say a is positive and if $a < 0$, we say a is negative.

Some properties of inequalities:

The following are consequences of the axioms which are provided without proofs.

Theorem 1.2. *If a, b and c are real numbers then*

1. $a < b$ *and* $c < 0$ *implies* $ac > bc$
2. $0 < 1$ *and* $-1 < 0$
3. $a > 0$ *implies* $1/a > 0$
4. $0 < a < b$ *implies* $0 < 1/b < 1/a$
5. $|a + b| \leq |a| + |b|$ *(triangle inequality)*
6. $||a| - |b|| \leq |a - b|$ *(a consequence of the triangle inequality)*

1.2.1 Order relation

We defined in $\mathbb{R}$ an order relation $\leqslant$ by $a \leqslant b$ or $b \geqslant a$. We recall the axioms:

$$(A_1) : \forall a \in \mathbb{R} : a \leqslant a \;\; (Reflexive) .$$

$$(A_2) : a \leqslant b \text{ and } a \leqslant b \Longleftrightarrow a = b \, (Antisymmetric) .$$

$$(A_3) : a \leqslant b \text{ and } b \leqslant c \Longrightarrow a \leqslant c \, (transitive) .$$

We can show that all elements of $\mathbb{R}$ are comparable with respect to the order relation $\leqslant$ and as such we imply a totally ordered set (total order relation)

1.2.2 Example

One can show the utility of such order relation

1. Let a be a real number such that $|a| < \varepsilon, \forall \varepsilon > 0$, we have $a = 0$ (elsewhere, if we choose $\varepsilon = \dfrac{|a|}{2}$, contradiction.

2. Let a and b be two real numbers such that $a < b + \varepsilon, \forall \varepsilon > 0$, then $a \leqslant b$ (else, if we take $\varepsilon = \dfrac{b-a}{2}$, we get a contradiction).

1.3 Natural Numbers and Induction

Definition 1.1. *Mathematical induction is a mathematical proof technique requiring essentially that a statement $P(n)$ holds for every natural number $n = 0, 1, 2, 3, ...$; that is, the overall statement is a sequence of infinitely many cases*
$P(0), P(1), P(2), P(3), ...$

A proof by induction consists of two steps: first step, the base, proves the statement for $n = 0$ without assuming any knowledge of other cases. Second step, the induction, proves that if a statement holds for any given case $n = k$, then it must also hold for the next case $n = k + 1$. These two steps establish that the statement holds for every natural number n. The base case doesn't necessarily begin with $n = 0$, but often with $n = 1$ and possibly with any fixed natural number $n = N$, establishing the truth of statement for all natural numbers $n \geqslant N$.

Example 1.1. *Prove the following statements:*

a- $1 + 3 + 5 + ... + (2n - 1) = n^2$

b- $1^2 + 2^2 + 3^2 + ... + n^2 = \frac{n(n+1)(2n+1)}{6}$

c- $1^3 + 2^3 + 3^3 + ... + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

Example 1.2. *Prove that for all numbers $x$different from 1:* $(1 + x)(1 + x^2)(1 + x^4)...(1 + x^2n) = \frac{1-x^2n+1}{1-x}$

1.4 Absolute value

The absolute value (or modulus) for any real number a, denoted by $|a|$, is defined as

$$a \longrightarrow |a| \in \mathbb{R}^+ = \begin{cases} a, \text{ si } a \geqslant 0 \\ -a, \text{ si } a < 0 \end{cases},$$

$$|a| = \sup(a, -a).$$

The absolute value has the following fundamental properties:

(1) $|a| = 0 \Longleftrightarrow a = 0.$

(2) $|a| = |-a|.$

(3) $|a.b| = |a| \cdot |b|$.

(4) si $a > 0 : |x| \leqslant a \Longleftrightarrow -a \leqslant x \leqslant a.$

Indeed, if $x \geqslant 0$ we have $x \geqslant -a$ and

$$|x| \leqslant a \Longleftrightarrow x \leqslant a.$$

if $x \leqslant 0$ we have $x \leqslant a$ and

$$|x| \leqslant a \Longleftrightarrow -x \leqslant a \text{ soit } x \geqslant -a.$$

(5) $|a+b| \leqslant |a| + |b|$.

Indeed, if a and b have the same sign, then the inequality is true. If $a \leqslant 0 \leqslant b$, then $a+b \leqslant b \leqslant b+|a|$ (because $a \leqslant 0 \leqslant |a|$), $|a| = -a$. i.e. $a+b \leqslant |a|+|b|$. also $b \geqslant 0 \geqslant -|b|$, $a+b \geqslant a \geqslant a-|b|$. which means that $a+b \geqslant -|a|-|b|$, and using (4), we get

$$|a+b| \leqslant |a| + |b|.$$

We can show, by induction that

$$|a_1 + a_2 + ... + a_n| \leqslant |a_1| + |a_2| + ... + |a_n|.$$

(6) $||a| - |b|| \leqslant |a-b|$.

$|a| = |b+(a-b)|$ and $|b| = |a+(b-a)|$, then, by (5), we obtain

$$|a| \leqslant |b| + |a-b|$$

$$|b| \leqslant |a| + |b-a|$$

$$-|a-b| \leqslant |a| - |b| \leqslant |a-b|,$$

by (4), we claim

$$||a| - |b|| \leqslant |a-b|.$$

1.5 Intervals

In mathematics, a (real) interval is a set that contains all real numbers lying between any two numbers. more precisely

Definition 1.2. *Let a and b bet two real numbers such that $b > a$. The set $\{x : a < x < b\}$ is called open interval and it noted by $]a, b[$. The set $[a, b] = \{x : a \leqslant x \leqslant b\}$ is called closed interval (compact interval). The sets $[a, b[= \{x : a \leqslant x < b\},]a, b] = \{x : a < x \leqslant b\}$, are called (respectively right and left) half-open intervals.*

For all intervals, the points a and b are called endpoints. If $a = b$, we put by definition $[a, a] = \{a\}$ (*degenerate closed interval*) and $]a, a[= \phi$. The length of the interval (closed, open, or half-open) is given by the real number $b - a$.

Examples: 1- The set $\{x : x \leqslant a\}$ is a left unbounded closed interval, noted $]-\infty, a]$. 2- The set $\{x : x < a\}$ is a left unbounded open interval, noted $]-\infty, a[$. 3- The set $\{x : x \geqslant a\}$ is a right unbounded closed interval, noted $[a, +\infty[$. 4- The set $\{x : x > a\}$ is a right unbounded open interval, noted $]a, +\infty[$. 5- The set $\mathbb{R}$ is also noted $]-\infty, +\infty[$. $-\infty$ and $+\infty$ represent infinity numbers.

1.6 Archimedean property

This property does not follow from the algebraic and order axioms alone. It states, informally, that there are no real numbers beyond all the natural numbers.

1.6.1 Archimedean axiom

For every real number a there is a natural number n such that $a < n$. Equivalently, the set $\mathbb{N}$ is not bounded above. We say that $\mathbb{R}$ is Archimidean.

Corollary 1.1. *For all real numbers a and b such that $a > 0$, there exists $n \in \mathbb{N}$ such that $na > b$.*

Proof. Just replace in the axiom a by $\frac{b}{a}$. □

Remark 1.1. *This property seems trivial, actually it's very important, and it allows us to define the famous definition of the integer part of a real number.*

Proposition 1.1. *Let $x \in \mathbb{R}$, there exists a unique integer (called integer part) denoted by $E(x)$, or $[x]$, such that:*

$$E(x) \leqslant x \leqslant E(x) + 1.$$

Example 1.3. $E(e) = 2$, $E(-e) = -3$, $E(1,45632) = 1$.

1.7 The completeness Axiom

In this section we give the completeness Axiom for $\mathbb{R}$. This Axiom will guarantee that $\mathbb{R}$ has no "gaps".

Definition 1.3. *Let S be a nonempty subset of $\mathbb{R}$.*

a. *If S contains the largest element s_0 [that is, s_0 belongs to S and $s \leqslant s_0$ for all $s \in S$, then we call s_0 the maximum of S and write $s_0 = \max S$.*

b. *If S contains the smallest element then we call the smallest element the minimum of S and write $\min S$.*

Example 1.4. *The set $\mathbb{R}$ has no maximum and minimum.*

Example 1.5. *The interval $]a, b[$ has no maximum nor minimum.*

Example 1.6. $\mathbb{N} = \{0, 1, ...\}$, $\min \mathbb{N} = 0$, $\mathbb{N}$ *has no maximum.*

Example 1.7. $\min\, [a, b[\,= a$, $\max\, [a, b[$ *does not exist.*

Example 1.8. *Let A be a subset of $\subset \mathbb{R}$, defined by $A = \{x \in \mathbb{R} : 0 \leqslant \ln x < 1\}$ Check the min and the max for A. Indeed, one has $0 \leqslant \ln x < 1$, which is equivalent to*

$$1 \leqslant x < e.$$

So $A = [1, e[$, we get, $\min A = 1$ and $\max A$ does not exist.

Definition 1.4. *Let S be a nonempty subset of $\mathbb{R}$.*

a. *If a real number M satisfies $s \leqslant M$ for all $s \in S$, then M is called an upper bound of S and the set S is said to be bounded above.*

b. *If a real number m satisfies $m \leqslant s$ for all $s \in S$, then m is called a lower bound of S and the set S is said to be bounded below.*

c. *The set S is said to be bounded if it is bounded above and bellow. Thus S is bounded if there exist real numbers m and M such that $S \in [m, M]$.*

Example 1.9. *The set $A = \{\sin n,\ n \in \mathbb{N}\}$ is bounded, because*

$$\forall n \in \mathbb{N} : -1 \leqslant \sin n \leqslant 1.$$

Example 1.10. *The set $A = [1, 3]$ is bounded, we can easily check that*

$$\forall x \in A : 1 \leqslant x \leqslant 3.$$

Example 1.11. *Let $A = \left\{\frac{1}{n+1},\ n \geqslant 1\right\}$. A given real $M \geqslant \frac{1}{2}$ is an upper bound for A, and a given real $m \leqslant 0$ is a lower bound for A.*

Definition 1.5. *Let S be a nonempty subset of $\mathbb{R}$*

a. *If S is bounded above and S has a least upper bound, then we will call it the supremum of S and denote it by $\sup S$.*

b. *If S is bounded below and S has a greatest lower bound, then we will call it the infimum of S and denote it by $\inf S$.*

Example 1.12. *Let* $A =]1, 2[$*, we have*

$$\forall x \in A : 1 \leqslant x \leqslant 2,$$

hence $\inf A = 1$, $\sup A = 2$.

Example 1.13. *For the set* $A = \left\{\frac{n-1}{n},\ n \geqslant 1\right\}$*, one has* $\inf A = 0$, $\sup A = 1$ *because*

$$\forall n \geqslant 1 : 0 \leqslant \frac{n-1}{n} \leqslant 1.$$

Example 1.14. *Let* $A = \left\{\frac{1}{n}, n = 1, 2, 3, 4\right\}$*, we have* $\inf A = \min A = \frac{1}{4}$ *and* $\max A = \sup A = 1$.

Theorem 1.3. *Every nonempty subset S of $\mathbb{R}$ that is bounded above has a least upper bound. In other words,* $\sup S$ *exists and is a real number.*

Corollary 1.2. *Every nonempty subset S of $\mathbb{R}$ that is bounded below has a greatest lower bound* $\inf S$*.*

Exercise 1.1. *Let A and B be nonempty subsets of real numbers such that $A \subset B$, prove that*

1. *If B is upper bounded, the* $\sup B$ *exists and* $\sup A \leq \sup B$*.*

2. *If B is lower bounded, the* $\inf B$ *exists and* $\inf B \leq \inf A$*.*

Exercise 1.2. *Let A and B be nonempty subsets of real numbers, show that*

1. $\sup(A \cup B) = \max \{\sup A, \sup B\}$.

2. $\inf(A \cup B) = \min \{\inf A, \inf B\}$.

3. *If $A \cap B \neq = \emptyset$, show that* $\sup(A \cap B) \leq \min \{\sup A, \sup B\}$ *and* $\inf(A \cap B) \geq \max \{\inf A, \inf B\}$*.*

1.8 Characterization of supremum and infimum

Theorem 1.4. *Let X be a subset of $\mathbb{R}$. The real M is the supremum for X, if and only if the following hold :*

a. $\forall x \in X$, *x satisfies* $x \leqslant M$.

b. $\forall \varepsilon > 0, \exists\ x_\varepsilon \in X$, *satisfying* $M - \varepsilon < x_\varepsilon \leq M$.

Remark 1.2. *(a) means that M is an upper bound for X. (b) indicates that $M - \varepsilon$ is not an upper bound for all $\varepsilon > 0$.*

Example 1.15. *Let be* $A = \left\{1 - \dfrac{1}{n}, n \geqslant 1\right\}$. *Prove that* $\sup A = 1$. *Using the upper bound characterization, one can see that 1)* $\forall x = 1 - \dfrac{1}{n} \in A : x = 1 - \dfrac{1}{n} \leqslant 1$. *2) We claim that :* $\forall \varepsilon > 0,\ \exists x_\varepsilon \in A : x_\varepsilon > 1 - \varepsilon$

$$\begin{aligned} x_\varepsilon &> 1 - \varepsilon \Leftrightarrow 1 - \frac{1}{n} > 1 - \varepsilon \\ &\Leftrightarrow \frac{1}{n} < \varepsilon \\ &\Leftrightarrow n > \frac{1}{\varepsilon}. \end{aligned}$$

So, let $\varepsilon > 0$ *and* $n \in N$ *such that* $n > \frac{1}{\varepsilon}$ *(n exists because $\mathbb{R}$ is Archimedean set, then*

$$x_\varepsilon = 1 - \frac{1}{n} > 1 - \varepsilon.$$

Thus $\forall \varepsilon > 0,\ \exists x_\varepsilon \in X : x_\varepsilon > 1 - \varepsilon$ *which means* $\sup A = 1$.

Theorem 1.5. *Let X be a subset of $\mathbb{R}$. The real m is the infimum for X, if and only if the following hold :*

a. $\forall x \in X$, *x satisfies* $x \geqslant m$.

b. $\forall \varepsilon > 0, \exists\ x_\varepsilon \in X$, *satisfying* $x_\varepsilon < m + \varepsilon$.

Example 1.16. *Let be* $A = \left\{1 + \dfrac{1}{n}, n \geqslant 1\right\}$. *Prove that* $\inf A = 1$.

We have 1) $\forall x \in A : x = 1 + \dfrac{1}{n} \geqslant 1$. *2) We prove that* $\forall \varepsilon > 0$, $\exists x_\varepsilon \in A : m \leq x_\varepsilon < 1 + \varepsilon$.*, indeed*

$$\begin{aligned} x_\varepsilon &< 1 + \varepsilon \Leftrightarrow 1 + \frac{1}{n} < 1 + \varepsilon \\ &\Leftrightarrow \frac{1}{n} < \varepsilon \\ &\Leftrightarrow n > \frac{1}{\varepsilon}. \end{aligned}$$

So, let be $\varepsilon > 0$ and $n \in N$ such that $n > \dfrac{1}{\varepsilon}$. Then $x_\varepsilon < 1 + \varepsilon$. Thus $\inf A = 1$.

Theorem 1.6. *Let* A, B *be two nonempty subsets of* $\mathbb{R}$. *Define*

$$A + B := \{x + y : x \in A \ \ \textit{and} \ \ y \in B\},$$

and

$$A - B := \{x - y : x \in A \ \ \textit{and} \ \ y \in B\}.$$

we have

$$\sup(A + B) = \sup A + \sup B \qquad \textit{and} \qquad \sup(A - B) = \sup A - \inf B.$$

Establish similar formulas for $\inf(A + B)$ *and* $\inf(A - B)$.

1.9 Extended real number line

Definition 1.6. *The extended real number line is obtained from the real number line* $\mathbb{R}$ *by adding two infinity elements* $+\infty$ *and* $-\infty$ *endowed by the totally order relation extended from that of* $\mathbb{R}$ *to* $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, *where* $\overline{\mathbb{R}}$ *denotes the extended real number line.*

Operations on $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ are defined by

$$\begin{aligned} x + (+\infty) &= +\infty + x = +\infty, \forall x \in \mathbb{R}, \\ x + (-\infty) &= -\infty + x = -\infty, \forall x \in \mathbb{R}, \end{aligned}$$

$$x\,(\pm\infty) = (\pm\infty)\,x = \begin{cases} \pm\infty \text{ si } x > 0 \\ \mp\infty \text{ si } x < 0 \end{cases},$$

$$\begin{aligned} (+\infty) + (+\infty) &= +\infty, \\ (-\infty) + (-\infty) &= -\infty, \\ (\pm\infty)\,(\pm\infty) &= +\infty, \\ (\pm\infty)\,(\mp\infty) &= -\infty. \end{aligned}$$

As the sum $(+\infty) + (-\infty)$ and the product $0\,(\pm\infty) = +\infty$ are not well defined, so $\overline{\mathbb{R}}$ does not have any algebraic structures.

1.10 Topology of the line $\mathbb{R}$

1.10.1 Open sets, closed sets, neighbourhood

Definition 1.7. *A subset A of $\mathbb{R}$ is said to be open if it is empty or if for every $x \in A$ there exists an open interval containing x and contained in A.*

In other words an open set in $\mathbb{R}$ is a set which is the union of open intervals. The following assertions are an almost immediate consequence of this definition.

$\mathbf{O}_1$ Every union (finite or infinite) of open sets is open;

$\mathbf{O}_2$ Every finite intersection of open sets is open;

$\mathbf{O}_3$ The line $\mathbb{R}$ and the empty set $\emptyset$ are open sets.

Property O_1 results from the fact that every union of sets, each of which is a union of open intervals , is itself a union of open intervals. To prove the property O_2, it is sufficient to prove it for the intersection of two open sets A, B : By hypothesis

$$A = \cup_i A_i, B = \cup_j B_j$$

where A_i and B_j are open intervals. Therefore

$$A \cap B = (\cup_i A_i) \cap (\cup_j B_j) = \cup_{i,j}(A_i \cap B_j).$$

Since each of the sets $A_i \cap B_j$ is either empty or an open interval, $A \cap B$ is open. Finally, property O_3 is obvious.

Example 1.17. *Every open interval is an open set.*

Example 1.18. *The union of the open interval* $]n, n+1[$ *where* $n \in \mathbb{Z}$, *is an open set.*

The intersection of infinite number of open sets is not always open. For example $\cap_{n\in\mathbb{N}^*}(\left]\frac{-1}{n}, \frac{1}{n}\right[) = \{0\}$.

Definition 1.8. *A subset* A *of* $\mathbb{R}$ *is said to be closed when its complement* $C_{\mathbb{R}}^{A}$ *is open.*

Each of the properties O_1, O_2, O_3 at one implies a dual property for closed sets.

Example 1.19. *Every closed interval* $[a, b]$ *(where* $a \leq b$*) is a closed set. Indeed, the complement of* $[a, b]$ *is the union of the two open intervals* $]-\infty, a[$ *and* $]b, +\infty[$, *and is therefore an open set.*

It should be observed that a set can be neither open nor closed.

Neighbourhood (or neighborhood): Let $x \in \mathbb{R}$ and $\varepsilon > 0$. A neighbourhood of x is a subset of $\mathbb{R}$ which contains an open interval $V_{(x,\varepsilon)} =]x - \varepsilon, x + \varepsilon[$, containing x.

Example 1.20. *The interval* $]-\varepsilon, \varepsilon[$ $(\varepsilon > 0)$ *is a neighbourhood of* 0. *The interval* $\left]-\frac{1}{n}, \frac{1}{n}\right[$ $(n > 0)$ *is a neighbourhood of* 0.

Properties

1) The intersection of finite neighbourhoods of a point x is also its neighbourhood
2) If x and y are two distinct real numbers of $\mathbb{R}$, there exist two neighbourhoods V of x and W of y such that $V \cap W = \emptyset$. ($\mathbb{R}$ is a separated space (or Hausdorff).

Proposition 1.2. *A subset* S *is open if and only if* S *a neighbourhood of all the points of* S.

Example 1.21. *If* $a, b \in \mathbb{R}$ $(a < b)$, *the intervals* $]a, b[\,,]-\infty, a[\,,]a, +\infty[$ *are neighbourhoods for all their points.*

Remark 1.3. $\mathbb{R}$ *is both open and closed subset, elsewhere* $[a, b[\,(a < b)$ *is neither open nor closed.*

The meaning which we have just given to the word "neighbourhood" appears different from the one defined in ordinary usage, since for us a point x of $\mathbb{R}$ has many neighbourhoods, and one of them is the space $\mathbb{R}$ itself.

Accumulation points of a set

Definition 1.9. *If A is a subset of $\mathbb{R}$, a point x of $\mathbb{R}$ is called an accumulation point of A if, in every neighbourhood of x, there exists at least one point of A different from x. In other words, if A is a subset of $\mathbb{R}$, a point x of $\mathbb{R}$ is called an accumulation point of A if, $\forall \varepsilon > 0$, $A \cap \left]x-\varepsilon, x+\varepsilon\right[/ \{x\} \neq \emptyset$.*

The set of accumulation points is denoted by A' (it can be empty).

Example 1.22. $A = [1, 2]$, *then* $A' = [1, 2]$.

Example 1.23. $A = \left]0, 1\right[\cup \{2\}$, *so* $A' = [0, 1]$.

Example 1.24. $A = \{1, 2, 3, 4\}$, *therefore* $A' = \phi$.

Example 1.25. $A = \left\{\frac{1}{n}, n \geqslant 1\right\}$, *thus* $A' = \{0\}$.

Remark 1.4. *An accumulation point of a set does not necessarily belong to the set. For example, the point 0 is an accumulation point of the set of point $A = \left\{\frac{1}{n}, n \geqslant 1\right\}$, but does not belong to this set. Again 0 and 1 are accumulation points of $\left]0, 1\right[$ without belonging to this interval.*

Proposition 1.3. *Every closed set contains its accumulation points. Conversely, every set which contains its accumulation points is closed.*

Proof. Let A be a closed set; if $x \in C_{\mathbb{R}}^{A}$, then the open set $C_{\mathbb{R}}^{A}$ is a neighbourhood of x and does not contain any point of A nor does it contain any point of A. Thus x cannot be an accumulation point of A. Conversely, if A is such that no point of $C_{\mathbb{R}}^{A}$ are an accumulation point of A, then there exists for each $x \in C_{\mathbb{R}}^{A}$ a neighbourhood of x not containing any point of A, and therefore contained in $C_{\mathbb{R}}^{A}$; the set $C_{\mathbb{R}}^{A}$ is thus a neighbourhood of each of its points, i.e., it is open; in other words, A is closed. $\square$

Isolated points

Definition 1.10. *An isolated point of a set A is a point x of A which is not an accumulation point of A. In other words, it is a point x of A which has a neighbourhood V such that $A \cap V = x$. $(\exists \varepsilon > 0,\ A \cap \,]x - \varepsilon, x + \varepsilon[\, = \{x\}$.*

Example 1.26. *Let $A = [0, 1] \cup \mathbb{N}$, then the isolated points of A are $\{2, 3, ..., n, ...\}$.*